

Finding the Rank of a Matrix: Pivot Method

The **rank** of a matrix, denoted as $\rho(A)$, is the number of linearly independent rows. We find this by converting the matrix into **Row Echelon Form** using a strict step-by-step algorithm.

The Golden Rule of Row Transformations

To avoid logical errors or "moving target" glitches, follow this one rule:

Only use a "Pivot Row" to change other rows if that Pivot Row itself is NOT being changed in the same step.

Step-by-Step Transformation Logic

Problem: Find the rank of Matrix A :

$$\begin{bmatrix} 1 & 4 & 9 & 16 \\ 4 & 9 & 16 & 25 \\ 9 & 16 & 25 & 36 \\ 16 & 25 & 36 & 49 \end{bmatrix}$$

Step 1: Use R_1 as the Anchor

We use the top-left unity (1) to clear the first column. R_1 remains frozen.

- $R_2 \rightarrow R_2 - 4R_1$
- $R_3 \rightarrow R_3 - 9R_1$
- $R_4 \rightarrow R_4 - 16R_1$

$$\begin{bmatrix} 1 & 4 & 9 & 16 \\ 4 & 9 & 16 & 25 \\ 9 & 16 & 25 & 36 \\ 16 & 25 & 36 & 49 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 4 & 9 & 16 \\ 0 & -7 & -20 & -39 \\ 0 & -20 & -56 & -108 \\ 0 & -39 & -108 & -207 \end{bmatrix}$$

Step 2: Simplify the Rows

To make the numbers easier, we can simplify R_2, R_3, R_4 independently. For example, dividing R_3 by 4:

- $R_3 \rightarrow R_3 \div 4$

$$\begin{bmatrix} 1 & 4 & 9 & 16 \\ 0 & -7 & -20 & -39 \\ 0 & -20 & -56 & -108 \\ 0 & -39 & -108 & -207 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 4 & 9 & 16 \\ 0 & -7 & -20 & -39 \\ 0 & -5 & -14 & -27 \\ 0 & -39 & -108 & -207 \end{bmatrix}$$

Step 3: Clear the Remaining Columns

By repeating the pivot process (using the new R_2 as the anchor to clear the rows below it), we eventually find that R_4 becomes a row of zeros because the original numbers followed a quadratic pattern (n^2), which always disappears after three layers of consistent subtraction.

Conceptual Breakdown

- Pivot Selection:** Always start with the a_{11} element. If it isn't 1, swap rows to make it 1.

2. **Elimination:** Use that 1 to create zeros in every cell below it.
3. **Repeat:** Move to the next diagonal element (a_{22}) and repeat the process for the remaining sub-matrix.
4. **Count:** The number of rows that contain at least one non-zero number at the end is your Rank.

Final Result

- **Rank of Matrix A:** $\rho(A) = 3$
- **Note:** For any $n \times n$ matrix of consecutive squares where $n \geq 4$, the rank will always be 3.